Teaching Experience and Course Overview

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Course Title: CSI 5350 - Programming Languages and Compilers.

Institution: Computer Science & Engineering, Oakland University.

Term/Year: Fall 2017, Fall 2018, Fall 2019, Fall 2020, Fall 2021.

Course Description: This interdisciplinary course bridges the fields of Compiler Techniques and Program Verification, focusing on advanced methods to ensure software correctness and safety. It equips students with the theoretical and practical tools needed to analyze, verify, and ensure the reliability of complex software systems, preparing them for both academic research and industry applications in software verification.

The course builds on traditional compiler techniques, such as data-flow analysis, control-flow analysis, type checking, and abstract interpretation, which form the foundation of modern program analysis. Students explore various approaches to program analysis and construct a compiler for a programming language, incorporating one of the techniques they have learned to analyze the program as part of their coursework.

Program analysis is increasingly relevant in industry because it enables the discovery of bugs and security vulnerabilities before the program is executed, through static analysis at compile time. Many platforms are adopting these techniques to automate the verification of third-party applications and ensure their safety on their platforms.

I have assisted in delivering this course numerous times, and the knowledge I gained through taking and assisting with the course has provided a foundation for my research. My responsibilities included grading assignments, running lab sessions, and helping students assimilate the material and complete their projects.

Role: Teaching Assistant / Instructor

Course Title: CSI 3430/APM 3430: Theory of Computation

Institution: Computer Science & Engineering, Oakland University.

Term/Year: Fall 2019, Fall 2020.

Course Description: This course introduces students to formal models of computation, exploring key concepts in automata theory, formal languages, and computability. Students will learn how different computational models—ranging from finite state automata to Turing machines—can recognize various classes of languages and address fundamental questions about what can and cannot be computed.

Throughout the course, students will study Finite Automata and Regular Languages, gaining the ability to design and analyze deterministic and non-deterministic finite state automata (DFA/NFA). They will also learn to identify and describe regular languages and represent them using regular expressions.

Additionally, Students also learn about a more powerful class of languages the context-free languages. These are the languages defined by context-free grammars and recognized by pushdown automata (PDAs). A PDA, or single-stack PDA, is essentially a finite-state machine equipped with a stack, which provides it with additional memory and greater expressive power. This added capability allows it to recognize languages that a finite-state automaton (FSA) alone cannot, such as those requiring balanced, recursive, or nested structures.

Students will also learn to explain the mechanics of a Turing machine and its variations. Furthermore, the course will explore the limits of computation through discussions of computability.

This course equips students with the foundational concepts of computation theory, laying the groundwork for understanding the capabilities and limitations of computer systems.

Role: Teaching Assistant / Instructor

Course Title: CSI 5610 - Advanced Data Structures and Algorithms

Institution: Computer Science & Engineering, Oakland University.

Term/Year: Fall 2019, Fall 2020.

Course Description: CSI 5610 provides an in-depth study of advanced data structures and the principles of algorithm design and analysis. The focus is on techniques for designing efficient algorithms using appropriate data structures, proving their correctness, and analyzing their computational complexity. Key topics include hash tables, data structures for combinatorial optimization, search trees, recurrence relations, divide-and-conquer strategies, dynamic programming, greedy algorithms, advanced graph algorithms, and linear programming. Additionally, the course incorporates algorithms from recent technical literature to explore current advancements in the field.

Role: Teaching Assistant / Instructor

Course Title: CS363 - Computer Organization.

Institution: Maharishi International University.

Term/Year: Winter 2021, Winter 2022.

Course Description: This course introduces the internal structure of computers, covering topics such as assembly language programming, machine language, and the design of digital logic circuits. Students study Boolean algebra and logic gates, as well as essential combinational and sequential circuits used in computer design. They learn the fundamentals of computer organization and architecture and gain hands-on experience programming in assembly language. Additionally, students explore key components of computer systems, including memory and the CPU, through topics such as flip-flops, decoders, multiplexers, registers, and sequential circuits.

Role: Lead Instructor

Course Title: CS-390: Fundamental Programming Practices

Institution: Maharishi International University

Term/Year: I have taught this course numerous times, including (but not limited to) Fall 2022, Spring 2023, and Fall 2023.

Course Description: The Fundamental Programming Practices (FPP) course provides students with a comprehensive understanding of key object-oriented programming principles, data structures, and algorithm development. The course uses Java as the object-oriented programming language and is designed to address any foundational gaps students may have as they begin their M.S. in Computer Science program, ensuring they acquire the essential programming and software engineering skills needed for advanced study.

Role: Lead Instructor.

Course Title: CS 401 - Modern Programming Practices

Institution: Maharishi International University

Term/Year: I have taught this course numerous times, including (but not limited to) Fall 2021, Winter 2022, Spring 2022, Fall 2023.

Course Description: The Modern Programming Practices (MPP) course builds on the foundational concepts covered in the FPP course, advancing students' proficiency in software development. This course equips students with advanced software engineering skills, preparing them to design, implement, and test complex software solutions using modern development practices. Students explore advanced object-oriented programming (OOP) concepts such as inheritance, interfaces, and polymorphism, and gain a solid understanding of object-oriented design principles, including the SOLID principles and the Open/Closed Principle, to write modular and maintainable code. The course also introduces common OOP design patterns to help students engineer robust, maintainable, and production-grade software systems. In addition, students are exposed to more sophisticated programming techniques using the Java programming language, including collection processing, lambdas and streams, annotations, Java generics, and multithreading.

Role: Lead Instructor

Course Title: CS489 - Applied Software Development

Institution: Maharishi International University.

Term/Year: Spring 2023

Course Description: CS489: Applied Software Engineering offers students a comprehensive exploration of the software development lifecycle, from conception to deployment, with a focus on enterprise-level solutions. This course emphasizes core principles, best practices, and modern tools and technologies for building robust, high-quality software. In this course, students learn how to analyze requirements, model designs, implement solutions using appropriate software architectures, test code, and deploy software to production environments. The course emphasizes real-world skills such as web development, microservices, and cloud-native application development, along with software security.

Role: Co-Instructor

Course Title: CS57x Practicums in Computer Science

Institution: Maharishi International University.

Term/Year: I have taught this course numerous times, including (but not limited to) Fall 2021, Winter 2022, Spring 2022, Fall 2023, Winter 2024.

Course Description: CS57x offers students a valuable opportunity to apply and expand their academic knowledge in a professional setting. As part of the MUM ComPro Master's program, the practicum involves practical work experience through CPT (Curricular Practical Training), allowing students to bridge theory and real-world applications. By the end of the course, students must submit a 1,000-word STAR report, summarizing their experience and achievements using the STAR (Situation, Task, Action, Result) framework. This report prepares students to articulate their professional experiences in future job interviews.

The instructor for the CS57x Practicum course is the student's faculty advisor from the Computer Science Department. Their role is to guide students through the practicum experience, ensuring that students apply their academic knowledge effectively in a professional setting. The instructor provides feedback and support throughout the course, helping students reflect on their practical experiences. Additionally, the instructor evaluates the final report, ensuring that students articulate their learning outcomes in alignment with the course's goals.

The course is graded on a Pass/No Pass (P/NP) basis, with the STAR report serving as the primary assessment. The STAR report, structured around Situation, Task, Action, and Result, is evaluated using a detailed rubric. The rubric assesses the completeness and clarity of each section, as well as how effectively students connect their practical experiences to the concepts and skills gained from their coursework. Each section is weighed differently, with particular emphasis on the actions taken and their outcomes. The instructor totals the points from the rubric to determine whether the student meets the criteria for a passing grade.

Role: Co-Instructor